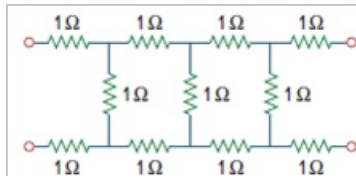


Problem

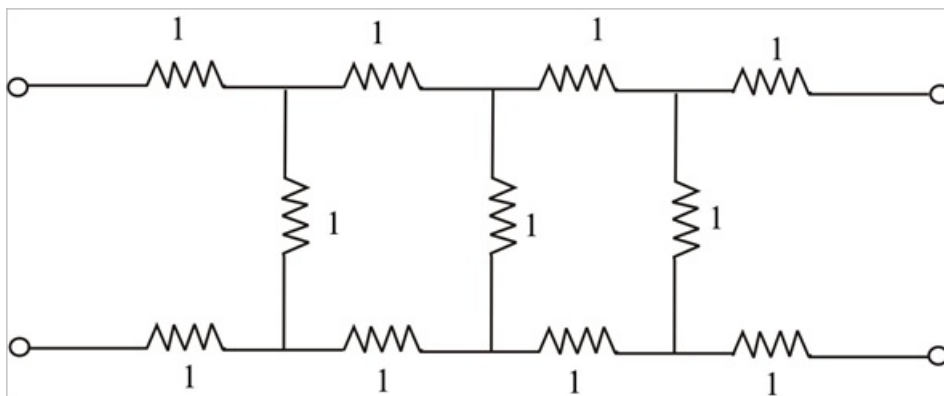
Find the impedance parameter equivalent of the network in Fig. 19.66.

Figure 19.66



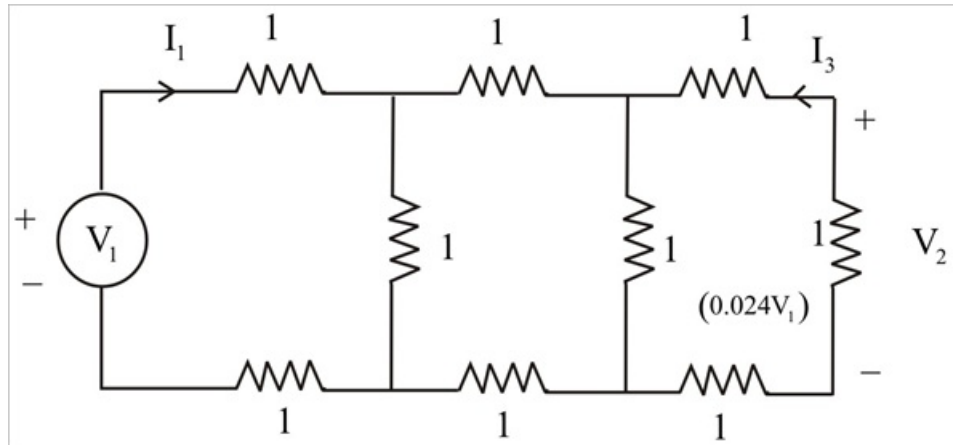
Step-by-step solution

Step 1 of 7

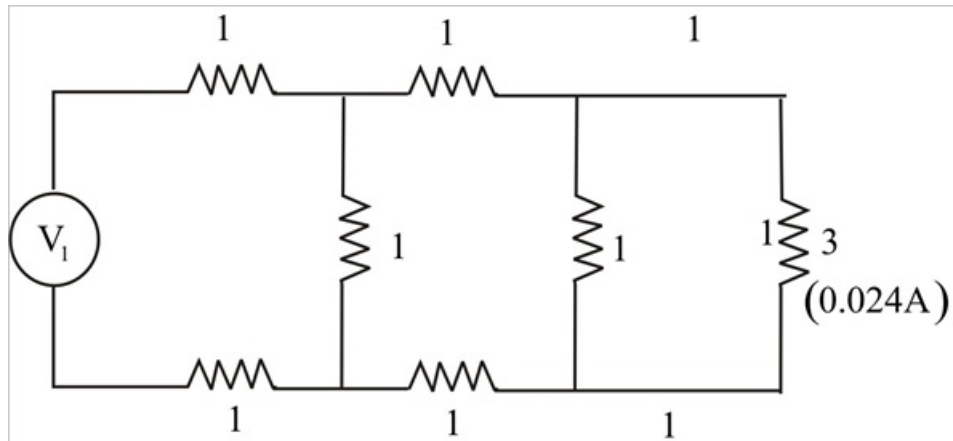


Step 2 of 7

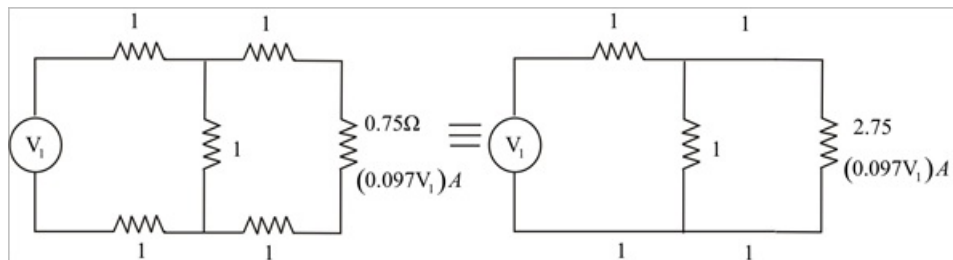
for $I_2 = 0$ we have :-



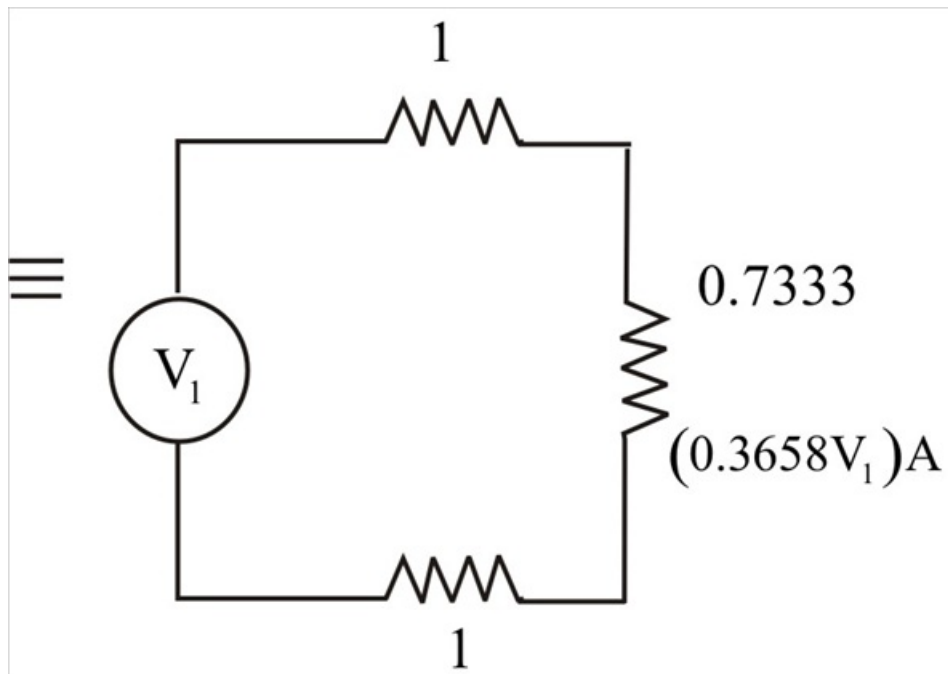
$$1113 = \frac{3}{4} = 0.75\Omega$$



Step 3 of 7



Step 4 of 7



Step 5 of 7

$$I_1 = \frac{V_1}{2.7333} = 0.3658V_1$$

$$Z_{11} = \frac{V_1}{I_1} \bigg|_{I_2=0} = 2.7333\Omega$$

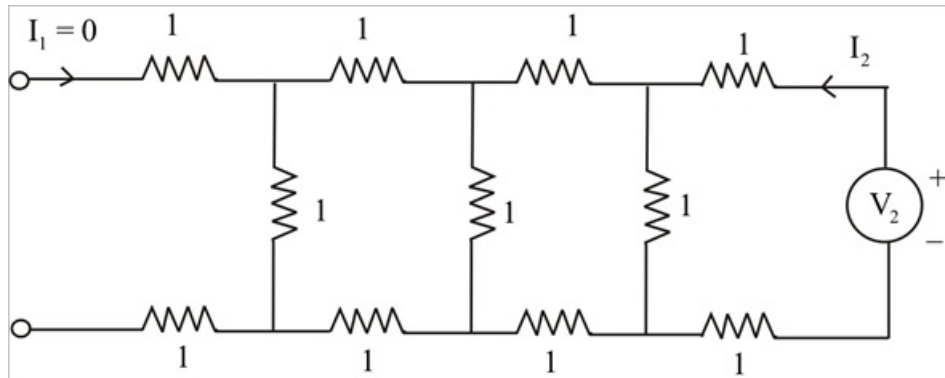
$$V_2 = 1 \times I_3, \quad V_2 = 0.0$$

$$Z_{11} = \frac{V_1}{I_1} \bigg|_{I_2=0} = 2.7333\Omega$$

$$Z_{21} = \frac{V_2}{I_1} \bigg|_{I_2=0} = 0\Omega$$

for $I_1 = 0$

Step 6 of 7



since it is a symmetrical ckt

$$Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0} = Z_{11}$$

$$Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0} = Z_{21}$$

$$Z_{22} = 2.733\Omega$$

$$Z_{12} = 0.067\Omega$$

$$Z = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 2.733 & 0.067 \\ 0.067 & 2.733 \end{bmatrix} \Omega$$

since it does not have any dependent sources and the network is linear and

$Z_{12} = Z_{21}$, this two-port is said to be reciprocal. Then the equivalent circuit becomes

Step 7 of 7

